

Student name:

CCNA v7.0 –SEMETER 3
FINAL SKILL EXAM
Duration: 1 hour 45 min

1. Object

- Build the network topology according to Device configuration and Physical connection tables.
- Configure LACP port-channel, VLAN, STP on routers/switched according to Planning section.
- Configure DHCP service, NTP, Syslog and TFTP as requirements.
- Configure OSPF routing protocol, static route and GRE tunnel to ensure connectivity between networks.
- Configure NAT and ACL for security requirements.

2. Device configuration

Location	Host name	Model
FPT Telecom	FPT_Telecom	Cisco CISCO1941/K9
FPT Telecom	FPT-SW	Cisco WS-C2960-24TT
VCB Ha Noi	HN-GW	Cisco CISCO2911/K9
VCB Ha Noi	HN-OFFICE	Cisco WS-C3560-24PS
VCB Ha Noi	HN-DATACENTER	Cisco WS-C3560-24PS
VCB Ha Noi	OFFICE-ACC1	Cisco WS-C2960-24TT
VCB Ha Noi	OFFICE-ACC2	Cisco WS-C2960-24TT
VCB Ha Noi	HN-DMZ	Cisco WS-C2960-24TT
VCB Ho Chi Minh	HN-GW	Cisco CISCO1941/K9
VCB Ho Chi Minh	HCM-SW	Cisco WS-C2960-24TT
FPT Telecom	fpt.vn	Server-PT
FPT Telecom	dns.fpt.vn	Server-PT
FPT Telecom	internet-user	Laptop-PT
VCB Ha Noi	vcb.com.vn	Server-PT
VCB Ha Noi	in.vcb.com.vn	Server-PT
VCB Ha Noi	dhcp.vcb.com.vn	Server-PT
VCB Ha Noi	NTP/Syslog/TFTP	Server-PT
VCB Ha Noi	IT_HN	PC-PT

Location	Host name	Model
VCB Ha Noi	ACC_HN	PC-PT
VCB Ha Noi	Guest_HN	Laptop-PT
VCB Ho Chi Minh	IT_HCM	PC-PT
VCB Ho Chi Minh	ACC_HCM	PC-PT
VCB Ho Chi Minh	Guest_HCM	Laptop-PT

3. Planning

3.1. Physical connection and port-channel

Host name	Interface	Port-channel	Port-channel	Interface	Host name	Note
FPT_Telecom	S0/0/0			S0/0/0	HN-GW	
FPT_Telecom	S0/0/1			S0/0/0	HCM-GW	
FPT_Telecom	Gi0/0			Gi0/1	FPT-SW	
FPT_Telecom	Gi0/1			NIC0	internet-user	
HN-GW	Gi0/0			Gi0/1	HN-DMZ	
HN-GW	Gi0/1			Gi0/1	HN-OFFICE	
HN-GW	Gi0/2			Gi0/1	HN-DATACENTER	
HN-OFFICE	Gi0/2			Gi0/2	HN-DATACENTER	
HCM-GW	Gi0/0			Gi0/1	HCM-SW	
HN-OFFICE	Fa0/21	Po1	Po1	Fa0/23	OFFICE-ACC1	Link Aggregation Protocol: LACP
HN-OFFICE	Fa0/22			Fa0/24	OFFICE-ACC1	
HN-OFFICE	Fa0/23	Po2	Po1	Fa0/23	OFFICE-ACC2	Link Aggregation Protocol: LACP
HN-OFFICE	Fa0/24			Fa0/24	OFFICE-ACC2	
FPT-SW	Fa0/1			NIC0	fpt.vn	
FPT-SW	Fa0/2			NIC0	dns.fpt.vn	
OFFICE-ACC1	Fa0/1			NIC0	IT_HN	
OFFICE-ACC1	Fa0/11			NIC0	Guest_HN	
OFFICE-ACC2	Fa0/6			NIC0	ACC_HN	
HN-DATACENTER	Fa0/1			NIC0	in.vcb.com.vn	
HN-DATACENTER	Fa0/2			NIC0	dhcp.vcb.com.vn	
HN-DATACENTER	Fa0/3			NIC0	NTP/Syslog/TFTP	
HN-DMZ	Fa0/1			NIC0	vcb.com.vn	
HCM-SW	Fa0/1			NIC0	IT_HN	
HCM-SW	Fa0/6			NIC0	ACC_HCM	
HCM-SW	Fa0/11			NIC0	Guest_HCM	

3.2. VLAN and Spanning-Tree

Location	VLAN ID	VLAN Name	STP Root primary	Access devices
VCB Ha Noi	10	Private_server	-	in.vcb.com.vn, dhcp.vcb.com.vn, NTP/Syslog/TFTP
VCB Ha Noi	110	IT	HN-OFFICE	IT_HN
VCB Ha Noi	120	ACC	HN-OFFICE	ACC_HN
VCB Ha Noi	150	Guest	HN-OFFICE	Guest_HN
VCB Ha Noi	190	Management	HN-OFFICE	-
VCB Ho Chi Minh	210	IT	-	IT_HCM
VCB Ho Chi Minh	220	ACC	-	ACC_HCM
VCB Ho Chi Minh	250	Guest	-	Guest_HCM

3.3. IP

Location	Link/Network	IP Network	Note
FPT Telecom	FPT LAN	222.222.222.0/24	
FPT Telecom	FPT – Internet user	111.111.111.0/24	
-	FPT_Telecom to HN-GW	200.0.0.0/30	
-	FPT_Telecom to HCM-GW	200.0.0.4/30	
VCB Ha Noi	HN-GW to HN-OFFICE	10.0.255.0/30	
VCB Ha Noi	HN-GW to HN-DATACENTER	10.0.255.4/30	
VCB Ha Noi	HN-OFFICE to HN-DATACENTER	10.0.255.8/30	
-	HN-GW to HCM-GW	10.0.255.12/30	GRE Tunnel
VCB Ha Noi	VLAN 110	172.16.110.0/24	
VCB Ha Noi	VLAN 120	172.16.120.0/24	
VCB Ha Noi	VLAN 150	172.16.150.0/24	
VCB Ha Noi	VLAN 190	172.16.190.0/24	
VCB Ha Noi	DATA CENTER LAN (VLAN 10)	172.16.10.0/24	
VCB Ha Noi	PUBLIC LAN	172.16.11.0/24	
VCB Ho Chi Minh	VLAN 210	172.16.210.0/24	
VCB Ho Chi Minh	VLAN 220	172.16.220.0/24	
VCB Ho Chi Minh	VLAN 250	172.16.250.0/24	

3.4. DHCP

VLAN ID	Device as DHCP Server	DHCP Configuration				
		Network	Default-router	DNS Server	Start IP	Number of IPs
110		172.16.110.0/24	HN-OFFICE	dns.fpt.vn	172.16.110.10	200

VLAN ID	Device as DHCP Server	DHCP Configuration				
		Network	Default-router	DNS Server	Start IP	Number of IPs
120	DHCP @ 172.16.10.5	172.16.120.0/24	HN-OFFICE	dns.fpt.vn	172.16.120.10	200
150		172.16.150.0/24	HN-OFFICE	dns.fpt.vn	172.16.150.10	200
210	HCM-GW	172.16.210.0/24	HCM-GW	dns.fpt.vn	172.16.210.10	200
220		172.16.220.0/24	HCM-GW	dns.fpt.vn	172.16.220.10	200
250		172.16.250.0/24	HCM-GW	dns.fpt.vn	172.16.250.10	200

3.5. IP Address assignment

Host name	Interface	IP Address	Subnet mask	Default Gateway	DNS Server
FPT_Telecom	S0/0/0	200.0.0.1	255.255.255.252	-	-
	S0/0/1	200.0.0.5	255.255.255.252	-	-
	Gi0/0	222.222.222.1	255.255.255.0	-	-
	Gi0/1	111.111.111.1	255.255.255.0	-	-
HN-GW	S0/0/0	200.0.0.2	255.255.255.252	-	-
	Gi0/0	172.16.11.1	255.255.255.0	-	-
	Gi0/1	10.0.255.1	255.255.255.252	-	-
	Gi0/2	10.0.255.5	255.255.255.252	-	-
	Tunnel0	10.0.255.13	255.255.255.252	-	-
HN-OFFICE	Gi0/1	10.0.255.2	255.255.255.252	-	-
	Gi0/2	10.0.255.9	255.255.255.252	-	-
	VLAN 110	172.16.110.1	255.255.255.0	-	-
	VLAN 120	172.16.120.1	255.255.255.0	-	-
	VLAN 150	172.16.150.1	255.255.255.0	-	-
	VLAN 190	172.16.190.1	255.255.255.0	-	-
HN-DATACENTER	Gi0/1	10.0.255.6	255.255.255.252	-	-
	Gi0/2	10.0.255.10	255.255.255.252	-	-
	VLAN 10	172.16.10.1	255.255.255.0	-	-
HCM-GW	S0/0/0	200.0.0.6	255.255.255.252	-	-
	Tunnel0	10.0.255.14	255.255.255.252	-	-
	Gi0/0.210	172.16.210.1	255.255.255.0	-	-
	Gi0/0.220	172.16.220.1	255.255.255.0	-	-
	Gi0/0.250	172.16.250.1	255.255.255.0	-	-
OFFICE-ACC1	VLAN 190	172.16.190.4	255.255.255.0	172.16.190.1	-
OFFICE-ACC2	VLAN 190	172.16.190.5	255.255.255.0	172.16.190.1	-

Host name	Interface	IP Address	Subnet mask	Default Gateway	DNS Server
fpt.vn	NIC0	222.222.222.222	255.255.255.0	222.222.222.1	-
dns.fpt.vn	NIC0	222.222.222.223	255.255.255.0	222.222.222.1	-
internet-user	NIC0	111.111.111.10	255.255.255.0	111.111.111.1	222.222.222.223
IT_HN, ACC_HN, Guest_HN, IT_HCM, ACC_HCM, Guest_HCM	NIC0	Dynamic Assignment			
in.vcb.com.vn	NIC0	172.16.10.4	255.255.255.0	172.16.10.1	222.222.222.223
dhcp.vcb.com.vn	NIC0	172.16.10.5	255.255.255.0	172.16.10.1	222.222.222.223
NTP/Syslog/TFTP	NIC0	172.16.10.6	255.255.255.0	172.16.10.1	222.222.222.223
vcb.com.vn	NIC0	172.16.11.10	255.255.255.0	172.16.11.1	222.222.222.223

3.6. NTP, Syslog and TFTP

- Routers and switches in VCB Ha Noi should be:
 - o Configured NTP and syslog.
 - o Backed up startup-configuration to TFTP server.
 - o Set enable secret, console and VTY password to [cisco123](#).

4. Routing

4.1. OSPF

- OSPF routing protocol is required to be enabled on HN-GW, HN-OFFICE and HN-DATACENTER to propagate VCB Ha Noi's IP networks.
- Default route should be propagated to another routers/switches by OSPF.

4.2. Static routing

Purpose	Network	Configuring router	Next-hop router	Note
Default route	0.0.0.0/0	HN-GW	FPT_Telecom	
Default route	0.0.0.0/0	HCM-GW	FPT_Telecom	
For Static NAT at HN-GW	200.200.10.0/28	FPT_Telecom	HN-GW	
For Dynamic NAT at HCM-GW	200.200.10.16/28	FPT_Telecom	HCM-GW	
For traffic between VCB HCM LAN networks and VCB Data Center network	172.16.10.0/24	HCM-GW	HN-GW	Over GRE tunnel
	172.16.210.0/24 172.16.220.0/24 172.16.250.0/24	HN-GW	HCM-GW	Over GRE tunnel
	172.16.210.0/24 172.16.220.0/24 172.16.250.0/24	HN-DATACENTER	HN-GW	

5. NAT

Type of NAT	Configuring router	Inside local	ACL name	Inside global	Pool name
Static NAT	HN-GW	172.16.11.10	-	200.200.10.1	-
Dynamic NAT	HCM-GW	172.16.210.0/24 172.16.220.0/24 172.16.250.0/24	HCM_TO_INTERNET	Pool IP of 200.200.10.16/28	HCM_POOL
PAT	HN-GW	172.16.110.0/24 172.16.120.0/24 172.16.150.0/24	HN_TO_INTERNET	S0/0/0 interface	-

6. Security requirements

6.1. Access-control lists

- FPT Telecom LAN network (*Internet*) should accept connections from Public IP only.
- Servers in VCB Data Center should accept connections from PCs in IT and ACC VLANs in both sites (*Ha Noi and Ho Chi Minh*) only (*Guest VLAN should be denied*).
- Server vcb.com.vn and in.vcb.com.vn should accept HTTP, HTTPS connections and ICMP only.
- Routers and switches in VCB Ha Noi should be remotely managed by PCs in IT VLAN only.

6.2. Other security requirements

- All routers and switches should be changed hostname according to Device configuration table.
- All in-use access ports on access switches should be configured STP PortFast and STP BPDU Guard feature.
- All not in-use ports on all switches should be shutted down.